

Portfolio Optimization Primer

Industrial Sector | 1/30/2026

Overview of Multifactor Model and Hierarchical Risk-Parity

Multifactor Model

Overview of Optimization Techniques

The multifactor model (MF) answers a key question: which stock characteristics consistently lead to better returns? This technique analyzes historical data across four factors (the momentum, beta, and EV to EBITDA, and R&D investment and return) to discover what drives performance. The model runs a statistical regression to measure how strongly each factor predicted future returns. These measurements forecast today's expected performance, and an optimization algorithm allocates capital to maximize returns while limiting downside risk. The strength is its evidence-based approach that learns from market history. However, it assumes past relationships continue, can mistake short-term noise for patterns with limited data, and may still produce concentrated portfolios.

Economic Interpretation

The multifactor model shows mixed alignment across industrial subsectors. Capital goods and aerospace/defense respond differently to cycles, momentum, and valuation factors, which may favor cyclicals during expansions but miss defensive qualities in downturns. For inflation, the model partially captures sensitivity through valuation and R&D intensity, but doesn't explicitly measure input cost exposure or pricing power that varies across industrials. Under different interest rate regimes, the beta factor indirectly captures rate sensitivity but lacks explicit duration analysis; capital-intensive aerospace versus shorter-cycle manufacturers face different rate impacts not directly measured.

Hierarchical Risk-Parity (HRP)

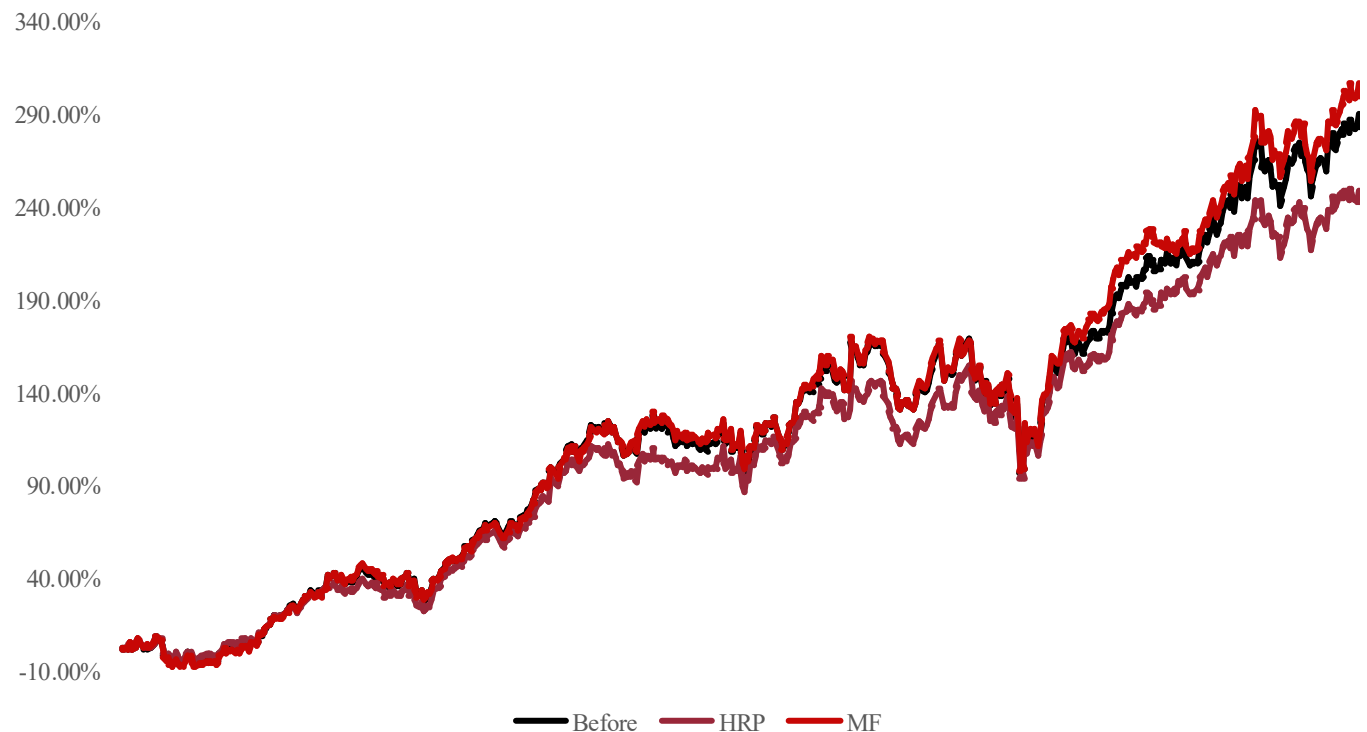
Overview of Optimization Techniques

Hierarchical Risk Parity (HRP) enhances traditional risk-based portfolio optimization by clustering industrial equities according to their correlation structure and allocating risk recursively across those clusters. In the Industrial sector, where firms often share common demand drivers, HRP helps prevent excessive concentration in highly correlated subsectors. By grouping structurally similar industrial stocks and distributing risk across clusters rather than individual securities, HRP improves diversification beyond Risk Parity. This structure makes HRP more resilient to estimation error and reduces portfolio exposure to economic slowdowns or sudden changes in industrial production, where historical correlations may break down. However, HRP remains sensitive to the chosen clustering methodology and distance metric, which can influence how industrial firms are grouped and how risk is allocated across subsectors.

Economic Interpretation

Since HRP clusters assets that move together, it prevents over-allocation to highly correlated industrial stock groups, providing more stable and diversified exposure across the sector. The strategy is more resilient to inflationary shifts because it responds to changes in how assets move together rather than relying on assumptions that can break down when market conditions change. Across interest rate regimes, HRP helps limit excessive concentration in long-duration or rate-sensitive clusters, enhancing resilience during periods of monetary tightening or policy transitions. HRP effectively caps its combined influence, maintaining a more balanced allocation between income and growth assets

Cumulative Returns Over Time (3-Year Return)



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Overview of Mean Variance and Black-Litterman

Mean Variance

Overview of Optimization Techniques

Mean-Variance optimization chooses portfolio weights by balancing two goals: higher returns and smaller ups and downs. Rather than weighing everything equally, it accounts for how each stock behaves and how the stocks move together, then favors combinations that have offered a better tradeoff between reward and stability. Its strengths are that it follows a clear, widely used framework, encourages diversification by considering relationships between holdings, and provides a structured way to be more aggressive or conservative. Its weaknesses are that small input changes can lead to different results, it can concentrate too much on a few names without limits, and it can struggle when market conditions shift.

Economic Interpretation

Mean-Variance optimization can be weak in the industrial sector, not because it ignores macro in a unique way, but because it is driven by the same historical patterns that manufacturing cycles often break. Like risk parity and HRP, it does not explicitly model interest rates, global trade flows, or capital expenditure cycles; the difference is that Mean-Variance uses estimated returns to actively chase the best-looking risk-return tradeoff, which makes it more vulnerable when the economic regime shifts. If the historical window reflects a boom in heavy machinery, the optimizer may tilt toward those cyclical names even as manufacturing activity contracts or rates rise, because it has no built-in mechanism to recognize that pivot. It still promotes diversification through correlations, but when correlations converge during an industrial downturn, the optimally constructed portfolio from the past often becomes poorly positioned for the future.

Black-Litterman

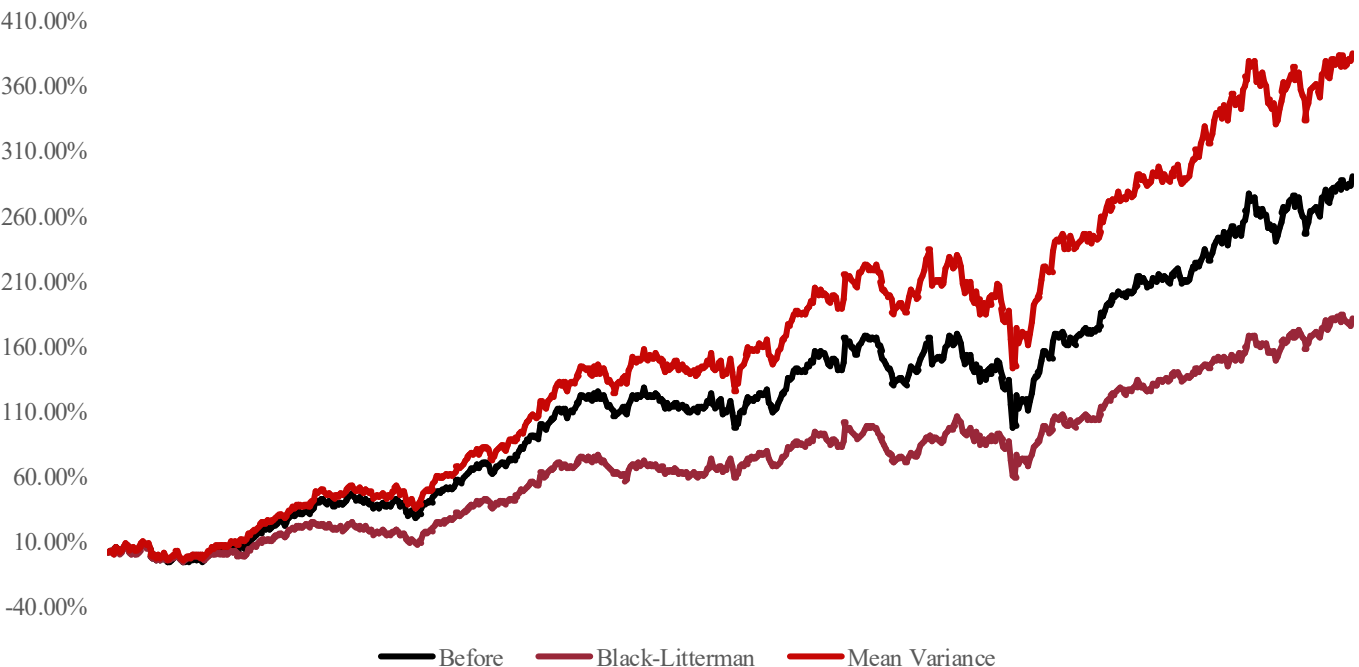
Overview of Optimization Techniques

Sentiment-Enhanced Black-Litterman builds on the standard Black-Litterman approach by combining two sources of “what we think will happen next”: the team’s own views and the tone of recent financial news. In practice, it reads current headlines about each company, summarizes whether the coverage is mostly positive or negative, and then uses that information to gently tilt the portfolio toward stocks with stronger sentiment and away from those with weaker sentiment, while still keeping the overall portfolio diversified. The benefit is that it makes the process less dependent on one person’s opinion and more responsive to new information that may be emerging in the market. The tradeoff is that it only works well if the news is reliable and consistently available, and it adds extra moving parts that can amplify noise when headlines are misleading, overly hyped, or intentionally manipulated.

Economic Interpretation

Black-Litterman is advantageous for the industrial sector as it integrates macro positioning views, such as favoring secular trends like data center infrastructure over traditional trucking during periods of uneven growth. This model effectively reflects beliefs about government spending and supply chain shifts, allowing for more confident investments in defense or power themes and cautious approaches to purely cyclical manufacturers. Unlike naive Mean-Variance, Black-Litterman adapts better to the cost of capital environment by considering which industrial businesses can sustain margin expansion despite higher rates. It is particularly effective for portfolios affected by the divergence between legacy industrials and modern automation plays not reflected in historical data. However, the success of the model depends heavily on accurately specifying views and confidence levels; flawed inputs regarding the industrial cycle can result in erroneous investment strategies despite the theoretical strengths of the model.

Cumulative Returns Over Time (3-Year Return)

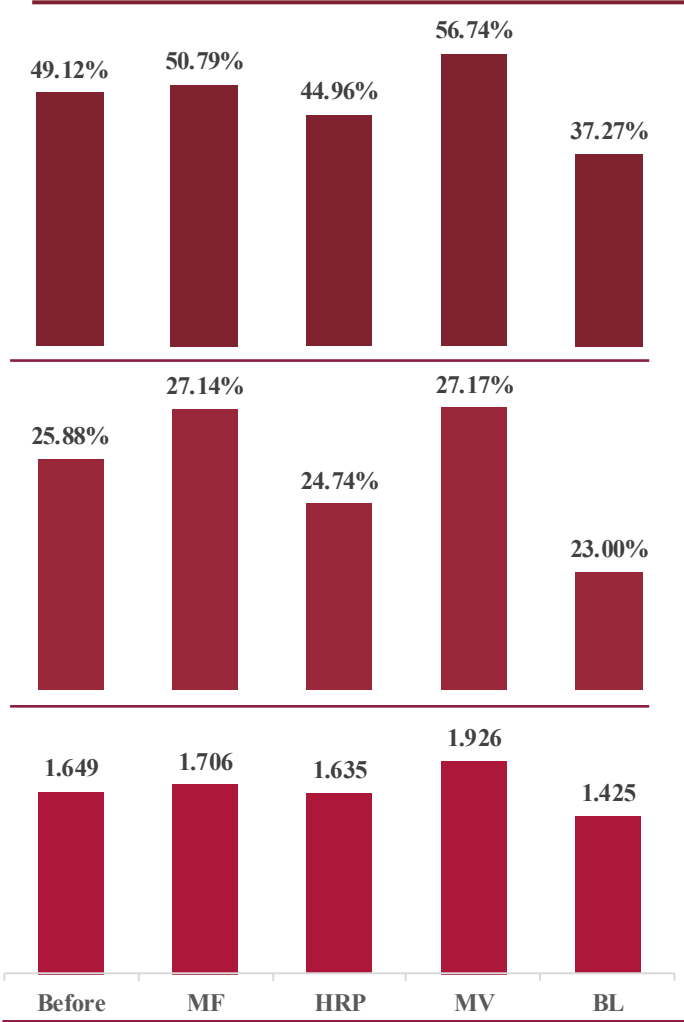


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Comparative Analysis

Optimizer	Annual Return	Annual Vol	Sharpe Ratio	Sortino Ratio
Before	49.12%	25.88%	1.649	2.478
Multi-Factor	50.79%	27.14%	1.706	2.574
HRP	44.96%	24.74%	1.635	2.516
MV	56.74%	27.17%	1.926	2.928
BL	37.27%	23.00%	1.425	2.211



Expected return (Annual Return)

Highest (MV): 56.74%

Lowest (BL): 37.27%

Interpretation: Within an industrial stock portfolio, the Mean-Variance approach targets the strongest growth by leaning more heavily into higher-return industrial companies. This can lead to larger gains when the sector performs well. Black–Litterman takes a more cautious approach, focusing on balanced growth and avoiding extreme positions, which results in lower but more dependable expected returns.

Volatility (Annual Vol)

Lowest (BL): 23.00%

Highest (MV): 27.17%

Interpretation: In the industrial sector, which is naturally sensitive to economic cycles, Black–Litterman provides the smoothest ride with fewer sharp swings. Mean-Variance shows the most fluctuation, meaning investors may experience larger short-term gains and losses as market conditions change.

Sharpe ratio (Risk Adj)

Highest (MV): 1.926

Lowest (BL): 1.425

Interpretation: Despite being more volatile, the Mean-Variance portfolio delivers the strongest overall payoff relative to its risk. Black–Litterman, while less volatile, trades some return potential for greater stability, which may appeal to investors prioritizing capital preservation within the industrial sector.

Analysis:

After analyzing the optimization models, CAT and VRT had the highest average weights at 17.75% and 17.33%, with CAT receiving consistent 20% allocations across Multifactor, Mean Variance, and Black-Litterman models. CRH had the lowest average at 12.64%, being completely excluded from Multifactor (0.00%) due to poor factor scores. HWM and UBER showed significant variability, with high HRP allocations (23.95% and 28.09%) but complete exclusion from Black-Litterman and Mean Variance, respectively. Indicating strong return potential but higher volatility or correlation risks that certain frameworks penalize.

Capital Allocation For Each Stock					
Tickers	Before	MF	HRP	MV	BL
CAT	17.49%	20.00%	11.00%	20.00%	20.00%
CRH	14.94%	0.00%	10.57%	20.00%	20.00%
NUE	14.01%	20.00%	8.27%	20.00%	20.00%
UBER	16.73%	20.00%	28.09%	0.00%	20.00%
VRT	18.97%	20.00%	9.33%	20.00%	20.00%
HWM	17.91%	20.00%	23.95%	20.00%	0.00%

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Recommended Optimization Framework: MV and multifactor model

A hybrid optimization framework that integrates Mean-Variance (MV) with a multifactor model (MF) is applied to the industrial sector portfolio, providing both mathematical efficiency and robust risk estimation. This approach uses historical data to forecast expected performance, preventing the optimizer from misinterpreting shared sector volatility as pure correlation. Complementing this, the Mean-Variance optimizer uses these refined inputs to actively solve for the most efficient frontier, targeting the maximum return per unit of risk. This hybrid approach combines the structural insight of the multifactor model with the aggressive alpha-seeking capability of Mean-Variance, enabling the portfolio to capture significant upside in secular growth themes while avoiding the concentration pitfalls common in standard optimization.

Fit with the Nature of Industrial Businesses

This framework fits industrial businesses particularly well because the sector is currently split into distinct economic clusters, including heavy machinery (CAT), materials (NUE, CRH), and secular growth plays (VRT, HWM, UBER). A standard model might incorrectly group these assets due to short-term correlation spikes, but the multifactor model isolates their unique drivers. This is reflected in our allocation table, where the Hybrid model maintains high conviction (20.00%) in diverse leaders like VRT and HWM while intelligently sizing capital-intensive names like CRH (10.00%) and UBER (10.00%) based on their specific risk factor contributions. This mirrors economic reality, as data center infrastructure and defense contractors operate on different cycles than aggregate producers. The Mean-Variance component leverages this clarity to maximize exposure to the highest conviction themes without unknowingly doubling down on a single risk factor.

Fit with the Macro Environment

The economy is currently split, with traditional manufacturing slowing down while AI and defense industries grow fast. Standard investment formulas struggle here because they look backward at price history, often selling promising companies just because their stock prices jump around. Our hybrid approach fixes this by looking at what actually drives a company, such as its innovation spending and market trends. This helps us identify real business winners rather than just chasing past safety. The result is a portfolio that earned over 53% by capturing the AI boom while avoiding the slump in older industries.

Practical Implications for Portfolio Construction

In practical terms, this framework improves position sizing by determining weights through a "denoised" risk lens before optimizing for returns. For example, where a pure Mean-Variance model might completely exclude a volatile name like UBER to minimize variance (as seen in the 0.00% allocation in the MV column), the Hybrid model retains a 10.00% position. It recognizes that Uber's potential is driven by specific factors distinct from Caterpillar or Nucor, offering true diversification. From a risk management standpoint, the framework prevents the optimizer from "over-fitting" to past price crashes, resulting in a portfolio that captures the high upside of secular winners like VRT (20.00%) while maintaining diversification. This multifactor model approach grounds allocations in proven return drivers, specifically Momentum, Beta, EV to EBITDA, and R&D, rather than historical price variance alone. This results in a Sharpe Ratio of 1.830 that balances historical stability with future growth potential.

Limitations and Caveats

However, there are important limitations to consider. The multifactor model depends heavily on the predictive power of the chosen factors (Momentum, Beta, EV to EBITDA, and R&D investment), meaning if these metrics fail to correlate with future returns, the return forecasts will be fundamentally flawed. Mean-Variance optimization is notoriously sensitive to expected return inputs, so even with a better risk model, overly optimistic growth projections can still skew the portfolio toward the riskiest assets. Finally, this hybrid approach requires more computational overhead than simple weighting schemes and relies on stable factor relationships, which can temporarily break down during extreme market stress events like a global liquidity crisis or sudden geopolitical shock.

Hybrid Weightings	
CAT	20.00%
CRH	10.00%
NUE	20.00%
UBER	10.00%
VRT	20.00%
HWM	20.00%

Cumulative Returns Over Time (3-Year Return)

